

Project on: Positional Artefacts Propagate Through Masked Language Model Embeddings

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Abstract

This report presents a partial reproduction of the paper *Positional Artefacts Propagate Through Masked Language Model Embeddings* (Luo, Kulmizev, and Mao 2021). The paper did not provide an implementation of their experiments so we decided to implement it and verify their results with slightly different conditions (due to our limited computing capacity). Using a reduced experimental setup based primarily on SST-2, we reproduce the presence of a single dominant outlier dimension across BERT layers, although its exact index differs from the one reported in the original paper. We also confirm through a position probing experiment that positional information is highly accessible in early layers and decreases in deeper ones. In addition, we show that clipping the dominant outlier dimension consistently reduces anisotropy and increases self-similarity, supporting the interpretation that this dimension acts as a structural bias in the representation space. However, unlike the original paper, our WiC experiment does not show an improvement after clipping, but only a slight decrease in accuracy, suggesting that the geometric effect of clipping is clearer than its downstream semantic benefit. Overall, our reproduction supports the main qualitative claims of the paper while highlighting the impact of implementation details and computational constraints on the quantitative results.

One interesting phenomenon observed in masked language models is the presence of *outlier dimensions* in the hidden representations. These dimensions exhibit unusually large values across many token embeddings and can dominate similarity computations in the embedding space. Previous studies have shown that contextualized representations produced by models such as BERT tend to be highly anisotropic, meaning that vectors occupy a narrow cone in the representation space rather than being uniformly distributed (Ethayarajh 2019).

The paper “*Positional Artefacts Propagate Through Masked Language Model Embeddings*” (Luo, Kulmizev, and Mao 2021) investigates the origin and role of these outlier neurons. The authors show that these dimensions are strongly related to positional information and that they propagate through transformer layers due to the Layer Normalization mechanism. Furthermore, they demonstrate that removing these outlier dimensions significantly affects the geometry of the representation space as well as certain downstream tasks.

In this project, we reproduce several of the main analyses from this paper. In particular, we focus on detecting outlier dimensions in BERT representations, studying their relationship with positional information through probing experiments, and analyzing the effect of clipping these dimensions on the geometry of the embedding space. Our goal is to evaluate whether the key observations reported in the original paper can be reproduced under a simplified experimental setup.

1 Introduction

Pretrained transformer-based language models have become the dominant paradigm in natural language processing (NLP) due to their ability to learn rich contextual representations from large corpora. Models such as BERT (Devlin et al. 2019) and RoBERTa (Liu 2019) achieved state-of-the-art performance during their releases on a wide range of tasks, including sentiment analysis, question answering, and natural language inference.

Despite their strong empirical performance, understanding the internal mechanisms of these models remains a challenging problem. In particular, recent work in Explainable Artificial Intelligence (XAI) has focused on analyzing the structure of the representations learned by transformer models in order to better understand how linguistic information is encoded (Belinkov and Glass 2020; Ethayarajh 2019).

2 Paper overview: understanding Outlier Dimensions in Transformer Representations

2.1 Contextual Representations and Anisotropy

Modern masked language models such as BERT (Devlin et al. 2019) and RoBERTa (Liu 2019) produce contextualized token representations whose geometry has been widely studied in recent years. Unlike static embeddings, these representations depend on the surrounding sentence context and are extracted from intermediate hidden states of deep transformer encoders.

Several works have shown that the vector spaces induced

by these models exhibit strong *anisotropy*, meaning that most token representations tend to concentrate in a narrow region of the embedding space rather than being uniformly distributed (Ethayarajh 2019). In practice, this leads to high cosine similarity even between randomly sampled token vectors. This phenomenon has important consequences for representation analysis, since similarity-based metrics may reflect structural properties of the embedding space rather than purely semantic information.

Understanding the sources of this anisotropy is therefore crucial for interpreting contextual representations. The paper studied in this project (Luo, Kulmizev, and Mao 2021) investigates a specific structural artifact in masked language model embeddings: the presence of extreme-valued dimensions, referred to as *outlier neurons*, which significantly affect the geometry of the representation space.

2.2 Outlier Dimensions in Masked Language Models

Luo et al. (Luo, Kulmizev, and Mao 2021) observe that contextualized representations produced by masked language models contain a small number of dimensions whose values consistently dominate the rest of the vector. When analyzing token embeddings extracted from models such as BERT and RoBERTa, the authors find that specific coordinates repeatedly correspond to the largest or smallest values across a large proportion of tokens and layers.

These dimensions therefore behave as *outliers* in the distribution of neuron activations. Empirical analyses on datasets such as SST-2 and QQP reveal that these extreme-valued coordinates appear systematically in the hidden representations of the models. Because these dimensions strongly influence vector norms and cosine similarities, they have a significant impact on the overall geometry of the embedding space and contribute to the anisotropic structure discussed previously.

2.3 Connection to Positional Information

To understand the origin of these outlier dimensions, Luo, Kulmizev, and Mao 2021 analyze the different components involved in the construction of the input representations of masked language models. In models such as BERT and RoBERTa, the initial token representation is obtained by summing token embeddings, segment embeddings, and positional embeddings. The authors observe that similar extreme-valued coordinates already appear in the positional embeddings themselves.

Further analysis shows that these coordinates are preserved and propagated through the network during the forward pass. In particular, the parameters of the Layer Normalization layers amplify specific dimensions associated with positional information. Because these normalization parameters are shared across tokens, the same dimensions remain dominant throughout the hidden layers of the model.

These observations suggest that the outlier neurons are closely related to positional information encoded in the representations. As a consequence, artifacts originating

from positional embeddings can propagate through the transformer encoder and influence the geometry of contextualized embeddings.

2.4 Methodology and Experimental Setup

To investigate the role of these outlier dimensions, Luo et al. (Luo, Kulmizev, and Mao 2021) conduct a series of empirical analyses on contextualized embeddings extracted from pretrained BERT and RoBERTa models. The authors first analyze token representations obtained from large corpora, using datasets such as SST-2 and QQP to collect a diverse set of sentences. By averaging token embeddings across layers and samples, they identify the dimensions that systematically correspond to extreme activation values.

To better understand the information encoded in these neurons, the paper introduces a neuron-level probing method. A linear probe is trained to predict the position of a token within a sentence based on its contextualized representation. Instead of relying solely on the probe weights, the contribution of each neuron is defined as the product between the neuron activation and the corresponding weight in the probe classifier. This allows the authors to identify which dimensions contribute most strongly to positional prediction.

Finally, the paper evaluates the impact of these outlier neurons by performing a simple intervention: the corresponding dimensions are set to zero in the contextualized embeddings, a process referred to as *clipping*. The resulting representations are then analyzed through several experiments, including measurements of embedding anisotropy, word sense discrimination using the WiC dataset, and sentence similarity evaluations. These experiments allow the authors to assess how removing the outlier dimensions affects the geometry and semantic properties of the representations.

3 Implementation

In this project, we focus on reproducing and analyzing part of this experimental framework. In particular, we investigate the presence of outlier dimensions in contextualized embeddings and study their impact on the geometry of the representation space. The following sections describe the implementation choices and experimental setup used to replicate these observations.

3.1 Experimental setup

All experiments in this work are conducted using the pretrained `bert-base-uncased` model available in the HuggingFace Transformers library. This model contains 12 transformer layers and produces contextualized representations of dimension 768 for each token (Devlin et al. 2019).

Throughout the experiments, BERT is used as a fixed encoder and is not fine-tuned. Instead, hidden representations are extracted from the model and analyzed in order to study the behavior of individual embedding dimensions.

For most experiments, we rely on the SST-2 dataset from the GLUE benchmark (Wang et al. 2019). For the word sense disambiguation experiment, we use the WiC (Word-in-Context) dataset (Pilehvar and Camacho-Collados 2019). This dataset contains pairs of sentences in which the same target word appears in different contexts, and the task is to determine whether the word has the same meaning in both sentences.

All experiments are implemented in Python using PyTorch and the HuggingFace Transformers library. Hidden representations are extracted layer by layer from BERT using a dedicated extraction pipeline and are subsequently analyzed using the procedures described in the following sections.

The code is structured with the core of the experiments and architectures in the `src` file, with a notebook that only runs the experiments and plots the results.

3.2 Hidden state extraction pipeline

In order to analyze the internal representations produced by BERT, we implement a pipeline that extracts hidden states for all tokens in a corpus of sentences. This pipeline is built around a wrapper class that handles tokenization, model execution, and the collection of hidden representations.

More specifically, input sentences are first tokenized using the HuggingFace tokenizer associated with the `bert-base-uncased` model. The tokenized inputs are then passed through the pretrained BERT model with the option `output_hidden_states=True`, which allows access to the representations produced at each layer of the transformer. For a given batch of sentences, this produces a list of hidden state tensors corresponding to the embedding layer and the twelve transformer layers of BERT.

The pipeline then processes these tensors to extract token-level representations that can be analyzed across the entire dataset.

To facilitate this analysis, we convert the batch representation into a flattened representation containing only valid tokens. Padding tokens are removed using the attention mask produced during tokenization. Optionally, special tokens such as `[CLS]` and `[SEP]` can also be filtered out depending on the experiment. After filtering, the remaining token embeddings are concatenated into a matrix of size (N, d) , where N denotes the number of valid tokens extracted from the batch.

This flattened representation allows us to aggregate token embeddings across the corpus and compute statistics at the level of individual dimensions. In particular, we use this representation to compute mean vectors for each layer, to identify outlier dimensions, and to construct datasets for the probing experiments described later in the report.

The extraction pipeline processes the corpus in batches in order to handle large datasets efficiently. For each batch of sentences, hidden states are collected from the model, filtered to remove invalid tokens, and appended to a global tensor containing embeddings for the entire dataset. This procedure allows us to analyze contextualized token rep-

resentations across all layers of the model while keeping the computational cost manageable.

The resulting hidden representations are then used in the subsequent experiments to study the behavior of outlier dimensions and their impact on the geometry of the representation space.

3.3 Outlier detection

Once token representations have been extracted from the model, the next step consists in identifying outlier dimensions in the hidden representations. Following the methodology described in the original paper, we analyze how often each embedding dimension corresponds to the minimum or maximum activation across tokens.

For a given transformer layer, we consider the matrix of contextualized token embeddings of shape (N, d) , where N denotes the number of valid tokens extracted from the corpus and d the embedding dimension. For each token embedding vector, we compute the indices of the dimensions corresponding to the smallest and largest activation values. This produces two sequences of indices representing, respectively, the dimensions that most frequently take extreme negative or positive values.

We then count how often each dimension appears as the argmin or argmax across all tokens in the dataset. If a particular dimension appears disproportionately often as the minimum or maximum activation, it can be interpreted as an outlier neuron according to the definition proposed by Luo, Kulmizev, and Mao 2021. The relative frequency of these events provides a simple statistic that allows us to quantify the dominance of specific dimensions in the representation space.

This procedure is repeated independently for each transformer layer. The resulting statistics include the most frequent argmin and argmax dimensions as well as the proportion of tokens for which these dimensions correspond to extreme values. By comparing these statistics across layers, we can determine whether the same dimensions consistently appear as outliers throughout the network.

These outlier statistics are subsequently used in several downstream experiments. In particular, the identified dimensions serve as candidates for the clipping intervention described in Section 3.5, where their impact on the geometry of the representation space and on several evaluation tasks is investigated.

3.4 Position probe

To investigate whether positional information is encoded in specific dimensions of BERT representations, we implement a probing experiment based on a simple linear classifier. The goal of this probe is to determine to what extent the position of a token in a sentence can be predicted directly from its contextualized embedding.

For a given transformer layer, we first construct a dataset of token representations paired with their corresponding positions in the input sequence. Using the hidden state extraction pipeline described in Section 3.2, we process the

corpus in batches and collect contextualized token embeddings. For each valid token, we record both its embedding vector and its position index in the tokenized sequence. Tokens whose position exceeds a predefined threshold are discarded in order to limit the number of prediction classes and keep the probing task tractable.

This procedure produces a dataset of pairs $(\mathbf{x}_i, y_i)$, where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the embedding of a token extracted from a given layer and y_i corresponds to its position in the sequence. In our implementation, the dimensionality of the embeddings is $d = 768$, which corresponds to the hidden size of the `bert-base-uncased` model. The resulting dataset is randomly shuffled and split into training, validation, and test subsets.

The probe itself is implemented as a linear classifier that maps token embeddings to position predictions. More precisely, we use a single linear layer without bias that projects the embedding vector to a vector of logits over the possible positions. The predicted position corresponds to the index of the largest logit. The model is trained using the cross-entropy loss and optimized with the Adam optimizer.

Training is performed for 5 epochs (10 for the paper of Luo, Kulmizev, and Mao 2021) on the training split, while the validation split is used to monitor the learning process. After training, the probe is evaluated on the held-out test set, and we report both classification accuracy and average loss. This procedure is repeated independently for several transformer layers in order to analyze how positional information evolves throughout the network.

This analysis allows us to determine whether the dimensions identified as outliers in Section 3.3 also play a dominant role in encoding positional information.

3.5 Clipping of outlier dimensions

In order to evaluate the impact of outlier neurons on the structure of BERT representations, we implement a simple intervention consisting of clipping the dimensions identified as dominant outliers in Section 3.3. The goal of this procedure is to remove the contribution of these dimensions and observe how the resulting representations behave in downstream analyses.

For each transformer layer, the outlier detection step produces statistics indicating which embedding dimensions most frequently correspond to the minimum or maximum activation values across tokens. These dominant dimensions are selected as candidate outlier neurons. In practice, the clipping operation consists of setting the values of these selected dimensions to zero in the embedding vectors. Given a token embedding $\mathbf{x} \in \mathbb{R}^d$, the clipped representation $\tilde{\mathbf{x}}$ is obtained by replacing the coordinates corresponding to the selected outlier dimensions with zero while leaving all other dimensions unchanged.

The clipping procedure is applied independently to each transformer layer using the outlier statistics computed previously. These modified representations are then used to evaluate how removing dominant dimensions affects several properties of the embedding space. In particular, we analyze the effect of clipping on the anisotropy of context-

tual embeddings, on the self-similarity of word representations, and on the performance of a word sense disambiguation task.

3.6 The effects of clipping the outlier dimension

To systematically understand the impact of this outlier dimension, the paper Luo, Kulmizev, and Mao 2021 design a series of experiments focusing on vector space anisotropy, the extent of contextualization (self-similarity), and the ability to differentiate word senses. The overarching motivation across these experiments is to assess how zeroing out (clipping) these outlier dimensions alters the fundamental geometric properties of the vector space and affects performance on intrinsic semantic tasks.

Contextualized word vectors have been observed to be highly anisotropic, meaning the average cosine similarity between uniformly randomly sampled words is unexpectedly close to 1. We hypothesize that the identified outlier neurons, which exhibit consistently extreme values compared to other dimensions, are the primary drivers of this narrow geometric cone. To verify this, we randomly sample sentences from the SST-2 training set to create sentence pairs. For each pair, we isolate a single random token from each sentence (discarding the rest) and compute their cosine similarity to measure the space’s anisotropy. We then evaluate the vectors again after clipping the outlier neurons by setting their values to zero, observing the geometric shift.

Beyond spatial clustering, we aim to measure the degree of contextualization using the self-similarity metric, which compares the internal representations of the same word across different contexts. We hypothesize that outliers artificially distort the perceived contextualization across layers. To investigate this, we sample unique words from the SST-2 training set that appear in different sentences. For a given word w appearing in n sentences, its average self-similarity in the l^{th} layer is computed as:

$$\text{SelfSim}(w) = \frac{1}{n(n-1)} \sum_{i \neq j} \cos(f_l^i(w), f_l^j(w))$$

where $f_l^i(w)$ and $f_l^j(w)$ are the internal representations of w in sentences i and j . We compute this metric before and after clipping the outliers, and subsequently adjust the self-similarity by subtracting it from each layer’s anisotropy measurement to isolate true contextual variation.

Building on the premise that clipping outliers harmonizes a word’s contextualized vectors, we hypothesize that this operation can explicitly aid in intrinsic semantic tasks, such as differentiating word senses. To test this, we utilize the word-in-context (WiC) dataset, a binary classification task determining if a target word shares the same meaning across two distinct sentences. Our procedure involves computing the cosine similarity between the contextualized vectors of the target word across the sentence pairs in the WiC training dataset. If this similarity exceeds a specified threshold, we assign a true label; otherwise, we assign a false label.

4 Results

Layer	Top min dim	Ratio	Top max dim	Ratio
1	308	79.69%	341	5.40%
2	308	97.22%	341	7.26%
3	308	96.53%	341	7.10%
4	308	97.55%	341	6.00%
5	308	98.50%	341	8.65%

Table 1: Most frequent argmin and argmax dimensions detected in BERT hidden representations.

4.1 Detecting outlier dimensions

Due to limited compute power, we reran the experiment with only the SST-2 dataset and limited the data to 5000 texts.

As we can see from figure 1, we found similar results to the original paper. More precisely, there is one outlier dimension that stood out as the minimum in at least 80% of the texts in all layers as we can see from table 1.

We notice that the specific outlier dimension is different (557th dimension in the original paper against 308th dimension in our case), which might be explained by the use of a different BERT (trained slightly differently) than the original paper (they did not specify which exact BERT implementation they used). However, the main point is the presence of exactly one outlier dimension, which is coherent.

Please note that the code can be found under Experiment 1 and Experiment 2 in the notebook, one for the figure and another to show the detail of which dimension (and at what frequency) is the outlier.

To prepare for future experiments, we made a clipped version of BERT in Experiment 3 of the notebook, and we notice that after clipping, the new "outliers" are only an extrema 20% of the time, which is much better.

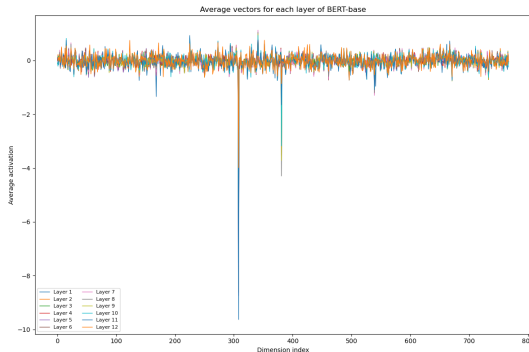


Figure 1: Average vectors for each layer of BERT-base.

4.2 Position probe results

Similarly to the previous subsection, we conduct this experiment with a cropped version of the dataset and with fewer epochs (5 instead of 10).

The results are noted in table 2. They are similar to what the authors obtained. The accuracy is high on the first

layers, and drops significantly for deeper layers. We notice a sharper drop in accuracy compared to the original paper.

The code can be found in Experiment 5 of the notebook.

Layer	Loss	Accuracy
1	0.0135	0.999
3	0.0605	0.995
6	0.6714	0.774
9	1.3186	0.519
12	1.9153	0.373

Table 2: Performance of the position probe across different BERT layers.

4.3 Effect of clipping on anisotropy

The results are reported in figure 2 and more quantitatively for the first few layers in table 3. We notice the same pattern as the ones shown in the original paper, namely that the anisotropy is dropping for every layer after we clipped the outlier dimension. However, the result is not nearly as sharp as the one in the original paper (in their implementation, the anisotropy almost dropped to zero when clipped). It could be due to a clipped dataset given as input.

The code is in Experiment 4 of the notebook.

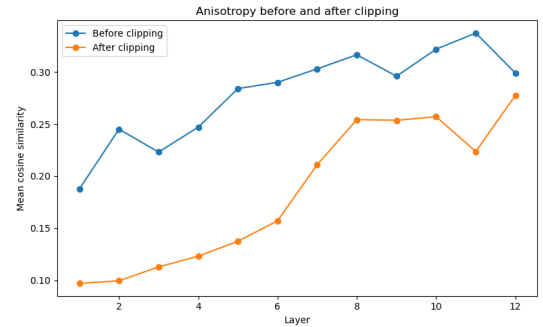


Figure 2: Anisotropy measurement of contextualized word vectors in BERT before and after clipping.

Layer	Before clipping	After clipping	Δ
1	0.1876	0.0969	-0.0906
2	0.2451	0.0994	-0.1457
3	0.2231	0.1128	-0.1103
4	0.2471	0.1231	-0.1240

Table 3: Effect of clipping the dominant outlier dimension on embedding anisotropy.

4.4 Effect of clipping on self-similarity

For this experiment, we chose to analyze 10 words instead of 1000 words in the paper, with the same threshold of appearance (10 times), throughout the entire SST-2 dataset. Moreover, the choice for which word is used for self-similarity is not detailed in the paper, therefore we decided to sample randomly words that fit the criteria given in the paper.

The results are in figure 3, we draw the same conclusion as in the paper, namely that clipping the outlier dimension increases the self-similarity. However, we notice that in the original paper, has a different trend in how the self-similarity move as we go deeper into the layers, which might be explained by the differences in the previous experiment. Because we need to account for the anisotropy when computing the self-similarity, we use the previous experiment’s result as the baseline to offset. The original paper having vastly different results in the previous experiment might explain the discrepancy in this experiment’s result.

The code can be found in Experiment 6 of the notebook.

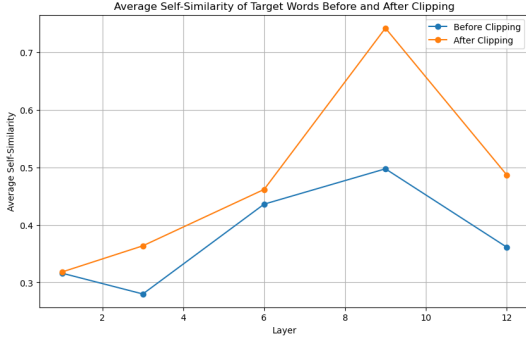


Figure 3: Self-similarity measurement of BERT before and after clipping.

4.5 Word sense disambiguation

The dataset WiC is a relatively small one, so we were able to use the same dataset as the original paper. However, due to limited computing power, we couldn’t test for every threshold possible to find the best like in the paper, so we did all our experiments with the threshold 0.7, but ran the experiment for every layer.

The result is in figure 4. We do not see the same results as in the paper. We notice that the accuracy slightly drops after clipping, which is not what the paper suggests (they highlighted the accuracy increasing after clipping).

Upon closer inspection, we can see that their table’s result is weak, as it compares the accuracy for different threshold and layer, moreover the increase in accuracy is negligible (less than 1% for BERT), which could be explained as noise.

As we will discuss in the qualitative analysis, such a small drop in accuracy is not necessarily inconsistent with the interpretation proposed by the paper. It rather suggests that removing the outlier dimension changes the geometry of the representation space much more clearly than it changes the semantic information useful for the task.

The code can be found under Experiment 7 of the notebook.

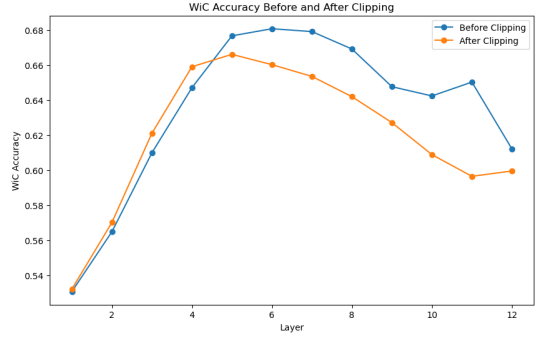


Figure 4: Accuracy scores on WiC dataset per layer for threshold 0.7 with BERT before and after clipping.

5 Qualitative Analysis

5.1 Role of positional information

A natural explanation for the emergence of an outlier dimension is that positional information is much more regular across samples than lexical or semantic information. While the meaning of tokens varies from sentence to sentence, every input sequence always contains a first token, a second token, and so on. Because this signal is present in a highly systematic way, it is plausible that one dimension of the hidden representation becomes strongly associated with this structural property and repeatedly takes extreme values across many tokens.

This interpretation is consistent with both the original paper and our reproduction. In our experiments, the dominant outlier dimension is not the same as in Luo et al. (Luo, Kulmizev, and Mao 2021): we observe dimension 308 whereas they report dimension 557. However, this does not contradict the main phenomenon. The exact index may vary with implementation details, initialization, or model checkpoints, while the qualitative pattern remains the same.

Our position probing experiment further supports this interpretation. The probe reaches almost perfect accuracy in the first layers and then degrades as depth increases (table 2), which indicates that positional information is strongly accessible in lower BERT layers. This matches the idea that the outlier dimension is not primarily semantic, but rather linked to a structural signal inherited from the input representation and preserved in the hidden states. From an explainability perspective, this is an important point: a highly salient neuron is not necessarily a neuron carrying interpretable semantic content. In this case, the dominant dimension appears to encode a systematic organizational property of the input rather than the meaning of the token itself.

5.2 Contribution of the outlier dimensions and the effect of clipping

Our experiments suggest that the main effect of the outlier dimension is to distort the geometry of the representation space more than to carry useful semantic information by itself. Before clipping, a large number of token vectors share one strongly dominant coordinate. As a consequence, vectors that are otherwise unrelated are par-

tially aligned along the same direction, which artificially increases their cosine similarity. This provides a natural explanation for the anisotropy observed in Figure 2: the representations are not close to each other because they express similar meanings, but because they all contain the same strong structural component.

This also explains why clipping reduces anisotropy in all layers. By setting the dominant coordinate to zero, we remove part of this common directional bias, so the vectors become less concentrated in a narrow cone. At the same time, anisotropy does not disappear completely in our reproduction. This is an important qualitative result: positional artifacts are clearly one source of anisotropy, but they are not the only one. Other global properties of contextualized embedding spaces, such as frequency effects or training-induced concentration, likely remain after clipping. This is also in line with the fact that our reduction in anisotropy is weaker than in the original paper.

The increase in self-similarity after clipping can be interpreted in the same way. Once the dominant structural coordinate is removed, two occurrences of the same word are compared less through a shared positional artefact and more through the rest of their representation. In other words, clipping removes a source of variation that is poorly related to lexical identity, which makes repeated occurrences of a word more comparable across contexts. The exact layer-wise trend differs from the original paper, but the qualitative effect remains the same in our experiments.

This perspective also helps explain the WiC results. In our reproduction, clipping slightly decreases accuracy instead of improving it. We do not view this as a strong contradiction. First, the original gain reported for BERT is already very small, so such a difference can easily be affected by experimental noise. Second, unlike the original paper, we evaluate WiC with a fixed threshold of 0.7, which makes the result more sensitive to the global scaling of cosine similarities before and after clipping. Third, removing an outlier dimension changes the geometry of the space, but this does not guarantee a monotonic improvement on every downstream comparison task. A dimension can be mostly structural and still remain weakly correlated with a useful decision boundary under a particular threshold.

Overall, our qualitative analysis supports the following interpretation: the dominant outlier dimension mainly acts as a global structural bias. It has a strong influence on a cosine-based measurements such as anisotropy, self-similarity, and similarity-based word sense evaluation, but it does not appear to encode the core semantic content of token representations. This is precisely why clipping produces a clear geometric effect while having only a limited impact on the practical word sense discrimination performance.

6 Project details

6.1 Limitations of the reproduction

There were no implementations at all from Luo, Kulmizev, and Mao 2021, so all the experiments and modules had to be coded by ourselves. We put great effort into implementing a structured and maintainable code base, allowing us

to easily add more experiments if needed while keeping the notebook clean.

Due to our limited computing power (one A4000 compared to four A100 from the authors), we weren't able to conduct the experiments with the scale, thus yielding different results, and even sometime contradicting ones.

For example, the pre-training part of the paper (compare the result of pre-training with and without positional embeddings) was not detailed and presented because it yield inconsistent results. We could only replicate it on a much smaller scale. We chose to do it on a much smaller dataset (`wikitext-103` instead of the `wikipedia` corpus) and with much fewer epochs (100 instead of 200 thousand steps for the original paper). However, it resulted in vastly different conclusions (the outlier is still present) to the paper which would undermine its main hypothesis. The code for this experiment can be found in Experiment 8 of the notebook.

6.2 Sensitivity to hyperparameters

Most of the experiments do not require specific hyperparameters since there is rarely training involved, moreover, the absence of an implementation from the author, or even some detail of their experiments (seed, batch size or learning rate for the probing) meant that our reimplementations probably used different hyperparameters, algorithms or even models (the outlier dimension being different hinted that we were using a different BERT from them), so it could be said that our reimplementations also served as a robustness test to their results. As discussed earlier, most of the experiments were robust to a cropped dataset, to fewer epochs. As a more detailed example, we can see from figure 5, which is a detailed plot of the experiment on self-similarity, with each word this time (instead of an average as in the paper). We can see the same pattern for most words, meaning that the self-similarity is increasing for the clipped version, further solidifying the robustness of the conclusion for this experiment.

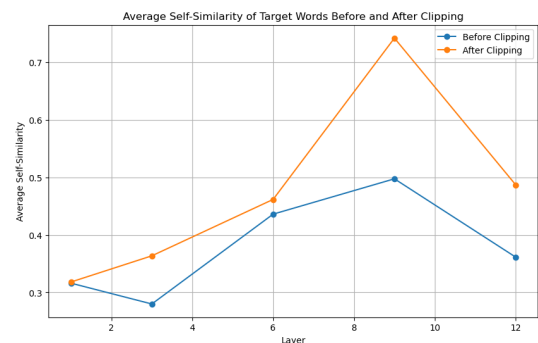


Figure 5: Self-similarity measurement of BERT before and after clipping for each word.

6.3 Use of LLMs

As requested in the course instructions, we explicitly describe how LLMs were used in this project.

LLMs were used for:

- clarifying the paper's methodology,
- debugging implementation details,

- improving the structure and wording of the report.

LLMs were not used to generate experimental results. All reported experiments were run by us, and all quantitative conclusions are based on our own code and outputs.

7 Conclusion and discussion

In this project, we reproduced the main qualitative phenomenon highlighted by Luo et al. (Luo, Kulmizev, and Mao 2021): BERT’s hidden representations contain a dominant outlier dimension that appears consistently across layers and is strongly related to positional information. Even though the exact coordinate we found differs from the one reported in the paper, the overall pattern remains the same, which suggests that the phenomenon is robust while the specific index of the outlier may depend on implementation details.

Our experiments also reproduce the main geometric effects described in the paper. In particular, clipping the dominant outlier dimension reduces anisotropy and increases self-similarity, which supports the idea that this dimension acts as a global structural bias in the representation space. The position probe results are also consistent with the original interpretation, as they show that positional information is highly accessible in the first layers and gradually fades in deeper ones. Taken together, these results support the central claim of the paper: some highly salient neurons in transformer representations reflect positional artefacts rather than semantic content.

At the same time, our reproduction also highlights several important nuances. First, the quantitative effects we obtain are generally weaker than those reported by the authors. This is especially visible for anisotropy, where clipping improves the geometry of the embedding space but does not make it nearly isotropic. Second, our WiC experiment does not show the small improvement reported in the paper, but rather a slight decrease in accuracy. We do not believe this weakens the main conclusion. Instead, it suggests that clipping mostly affects cosine-based ge-

ometry and similarity measurements, while its effect on downstream semantic discrimination is more limited and sensitive to evaluation choices such as threshold selection.

Overall, despite computational constraints and some quantitative discrepancies, our reproduction supports the main interpretation proposed by Luo, Kulmizev, and Mao 2021. It shows that analyzing individual dimensions of hidden states can reveal important biases in representation geometry, and that these biases can strongly affect common similarity-based analysis. Future work could extend this study in two directions: first, by reproducing the full set of experiments under closer computational conditions to the original paper, and second, by investigating whether similar positional artifacts arise in other architectures or in multimodal transformer models.

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